

- Developed by “Debuggers”

Zombie Survival Game

OUR TEAM CONSISTS

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Project Introduction

- **Zombie Survival Game**
- A survival-based game developed using C#
- Player acts as a survivor
- Fight zombies using different weapons
- Collect useful items
- Manage health and inventory
- Gain XP and level up
- Main goal: Survive zombie waves

Main Objectives

- ❑ Apply C# OOP concepts in a practical project
- ❑ Create classes and objects
- ❑ Implement inheritance and polymorphism
- ❑ Implement constructors and overloading
- ❑ Develop a combat system
- ❑ Manage player health and inventory
- ❑ Implement XP and level system
- ❑ Create a complete playable game

Main Classes

- **Player** — Represents the survivor
- **Zombie** — Represents enemies
- **Weapon** — Handles weapons
- **Item** — Food, medicine and useful items
- **Inventory** — Manages items and weapons
- **Shelter** — Represents safe place
- **Zombie Game** — Controls the overall game

Player Feature

Health
Energy
XP
Level
Attack

This is to manage
the player's status
and actions.

Zombie Feature

Health
Damage
Attack

This is to create enemies and
combat.

Weapon Feature

Name
Damage
Attack functionality

This is to provide different ways to
attack zombies.

Item System

- I. Food
- II. Medicine
- III. Useful items

Purpose: Help the player survive.

Inventory System

- i. Add items
- ii. Remove items
- iii. Store weapons
- iv. Store useful items
- v. Capacity management

Purpose: Manage collected resources.

Shelter

- Safe location
- Player protection
- Recovery/management

Purpose: Give the player a safe place.

OOP Concepts Used

- ☐ Classes & Objects
- ☐ Encapsulation
- ☐ Constructors
- ☐ Constructor Overloading
- ☐ Method Overloading
- ☐ Inheritance
- ☐ Abstract Class
- ☐ Interface
- ☐ Virtual Method
- ☐ Method Overriding
- ☐ Static Members
- ☐ Copy Constructor
- ☐ Operator Overloading



This will continue until you choose to exit this game.

Feature	This is Added For	Working Process
Combat	Fight zombies	Player uses weapons
Health	Track survival	Damage decreases health
Inventory	Manage resources	Add/remove items
Weapons	Different attacks	Each weapon has damage
Items	Help survival	Food/medicine used
XP	Progression	Gain XP after defeating zombies
Level	Player progression	Level increases with XP
Shelter	Safe place	Player can recover/manage
Zombie Waves	Increase challenge	New enemies appear

Project Output

- ❖ Game successfully runs
- ❖ Player selection works
- ❖ Combat system works
- ❖ Weapons can be used
- ❖ Zombies can be attacked
- ❖ Items can be collected
- ❖ Inventory works
- ❖ Health system works
- ❖ XP/Level system works
- ❖ Victory/Game Over works

Key Findings

- I. OOP helped organize the project
- II. Classes separated different responsibilities
- III. Inheritance reduced code duplication
- IV. Polymorphism improved flexibility
- V. Inventory added resource management
- VI. Combat system made the game interactive
- VII. XP system added player progression

Challenges

- i. Managing multiple classes
- ii. Connecting different classes
- iii. Managing player/zombie health
- iv. Maintaining game flow
- v. Implementing OOP concepts correctly

Conclusion

- Successfully developed Zombie Survival Game
- Used C# and OOP
- Implemented combat system
- Implemented inventory & item system
- Implemented health & XP system
- Improved understanding of OOP

Future Scope

- GUI
- More zombie types
- More weapons
- Sound effects
- Animations
- Multiplayer
- Save/Load system



Thank You